

Project Title

Library Book Request in ServiceNow

Name : Balaga Pravalika

College : Aditya Institute of Technology and Management

Email : 23a51a0506@adityatekkali.edu.in

Project Report

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Maximum Marks	

Project Overview

The **Library Book Request Management System** is developed using the ServiceNow platform to automate the process of requesting books from a college library. Traditionally, students submit requests manually, making it difficult for librarians to track requests and manage records efficiently.

This project provides an automated solution where students can submit book requests through a custom form. The system validates the Issue Date, automatically generates a unique Request Number, stores the request in a custom table, and sends an email notification to the librarian whenever a new request is submitted

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1. Introduction

College libraries handle numerous book requests every day. Manual request handling often causes delays, duplicate entries, and difficulty in maintaining records.

The Library Book Request Management System automates the complete request process. Students submit requests through a ServiceNow form, while the system validates the Issue Date, generates a Request Number automatically, stores request information, and notifies the librarian through email.

2. Project Objectives

The objectives of this project are:

- Develop a custom Library Book Request application.
- Create a custom table for storing requests.
- Validate that the Issue Date is not in the past.
- Generate Request Numbers automatically.
- Notify the librarian through email.
- Reduce manual work.
- Improve request tracking.
- Maintain library request records.

3. Key Features

- Custom Library Book Request Table
- Student Request Form
- Automatic Request Number Generation
- Issue Date Validation
- Email Notification to Librarian
- Centralized Record Management
- Easy Search and Tracking
- Secure Data Storage

4. Project Milestones

Milestone 1 – Requirement Analysis

Activities

- Identify project requirements
- Design database structure
- Identify required fields

Milestone 2 – Custom Table Creation

Activities

- Create Library Book Request table
- Add required fields
- Design the request form

Milestone 3 – Business Rule Development

Activities

- Generate Request Number automatically
- Validate Issue Date
- Prevent invalid requests

Milestone 4 – Notification Configuration

Activities

- Configure Email Notification
- Configure recipient
- Test notification delivery

Milestone 5 – Testing

Activities

- Validate records
- Test Business Rules
- Test Email Notifications
- Fix identified errors

5. ServiceNow Developer Setup

The development environment was prepared using the ServiceNow Developer Program.

Steps

1. Create a ServiceNow Developer Account.
2. Request a Personal Developer Instance (PDI).
3. Log in to the ServiceNow instance.
4. Access the Global Application Scope.
5. Begin application development.

6. Project Implementation

6.1 Create Custom Table

Navigate to:

System Definition → Tables

Create a table named:

Library Book Request

Fields

Field	Type
-------	------

Request Number	String
----------------	--------

Student Name	Reference (User)
--------------	------------------

Book Name	String
Author	String

Issue Date	Date
------------	------

Status	Choice
--------	--------

6.2 Form Design

The request form contains the following fields:

- Student Name
- Book Name
- Author
- Issue Date
- Status

6.3 Business Rules

Business Rule 1 – Generate Request Number

Automatically generates unique request numbers such as:

- LIB1001
- LIB1002
- LIB1003

Business Rule 2 – Validate Issue Date

Validates that the selected Issue Date is not earlier than the current date.

If a past date is entered, the system displays the message:

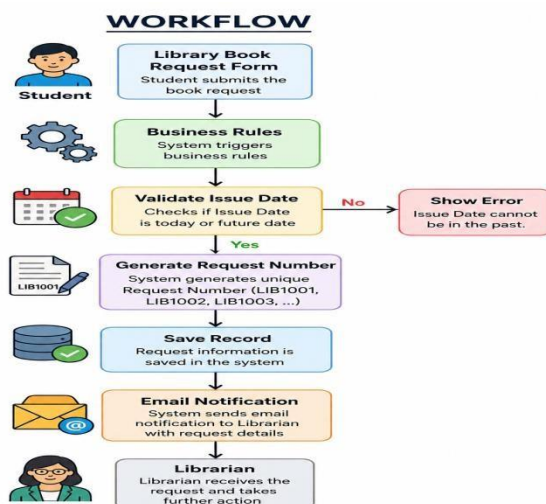
"Issue Date cannot be in the past."

6.4 Email Notification

Whenever a student submits a request:

- An automatic email is sent to the librarian.
- The email contains: o Request Number o Student Name o Book Name o Issue Date o Status

Workflow



7. Screenshots

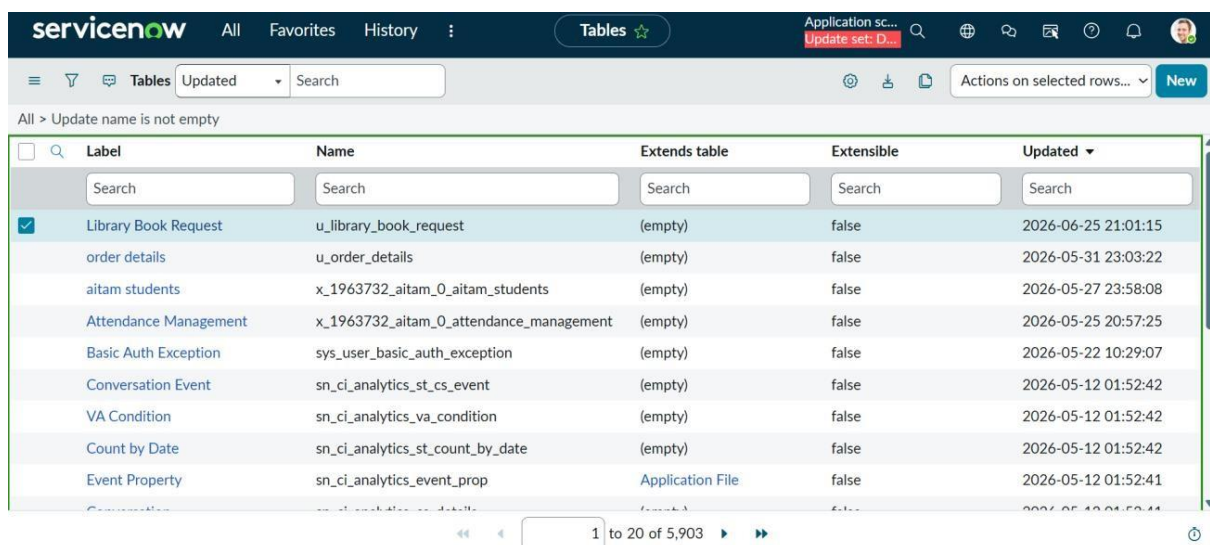
Screenshot 1: Custom Table

Creating the Custom Table

The custom table **Library Book Request** was created to store all book request information submitted by students. This table serves as the central repository for managing library book requests.

Steps:

1. Navigate to **System Definition** → **Tables**.
2. Click **New**.
3. Enter **Library Book Request** as the table label.
4. The table name is generated automatically.
5. Click **Submit** to create the table.



The screenshot shows the ServiceNow 'Tables' page. The table list includes the following data:

Label	Name	Extends table	Extensible	Updated
Library Book Request	u_library_book_request	(empty)	false	2026-06-25 21:01:15
order details	u_order_details	(empty)	false	2026-05-31 23:03:22
aitam students	x_1963732_aitam_0_aitam_students	(empty)	false	2026-05-27 23:58:08
Attendance Management	x_1963732_aitam_0_attendance_management	(empty)	false	2026-05-25 20:57:25
Basic Auth Exception	sys_user_basic_auth_exception	(empty)	false	2026-05-22 10:29:07
Conversation Event	sn_ci_analytics_st_cs_event	(empty)	false	2026-05-12 01:52:42
VA Condition	sn_ci_analytics_va_condition	(empty)	false	2026-05-12 01:52:42
Count by Date	sn_ci_analytics_st_count_by_date	(empty)	false	2026-05-12 01:52:42
Event Property	sn_ci_analytics_event_prop	Application File	false	2026-05-12 01:52:41

Screenshot 2: Request Form

Designing the Request Form

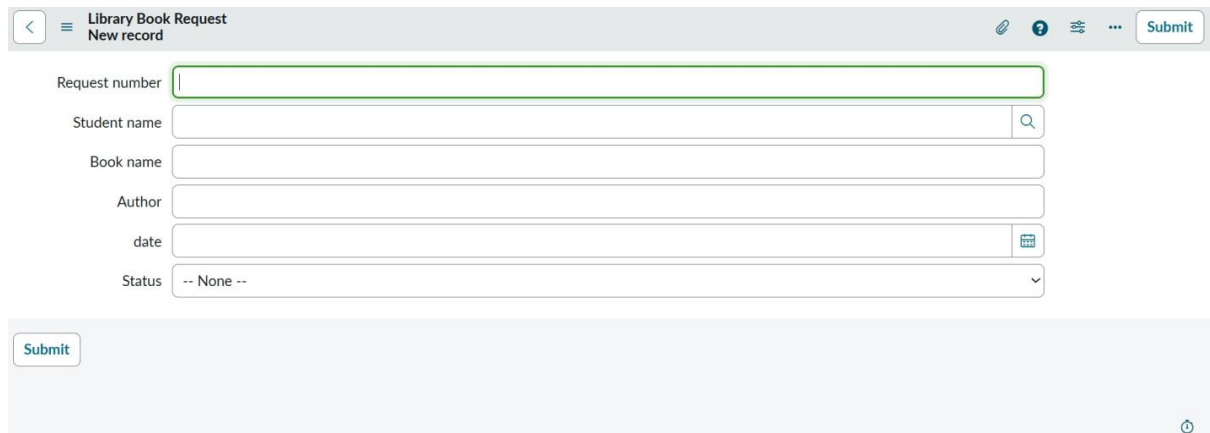
A user-friendly form was designed to allow students to submit book requests.

The form includes the following fields:

- Request Number
- Student Name
- Book Name
- Author
- Issue Date
- Status

Steps:

1. Open the Library Book Request table.
2. Right-click the form header.
3. Select **Configure** → **Form Layout**.
4. Add the required fields to the form.
5. Save the layout.



The screenshot shows a web application interface for configuring a form layout. At the top, there is a header bar with a back arrow, a menu icon, the text "Library Book Request New record", and icons for edit, help, and settings, followed by a "Submit" button. Below the header, the form layout is displayed with the following fields: "Request number" (a text input field with a green border), "Student name" (a text input field with a search icon), "Book name" (a text input field), "Author" (a text input field), "date" (a date picker field), and "Status" (a dropdown menu showing "-- None --"). At the bottom left, there is a "Submit" button, and at the bottom right, there is a refresh icon.

Screenshot 3: Generate Request Number Business Rule

Creating the Generate Request Number Business Rule

A Business Rule was created to automatically generate a unique request number whenever a new record is inserted.

Example:

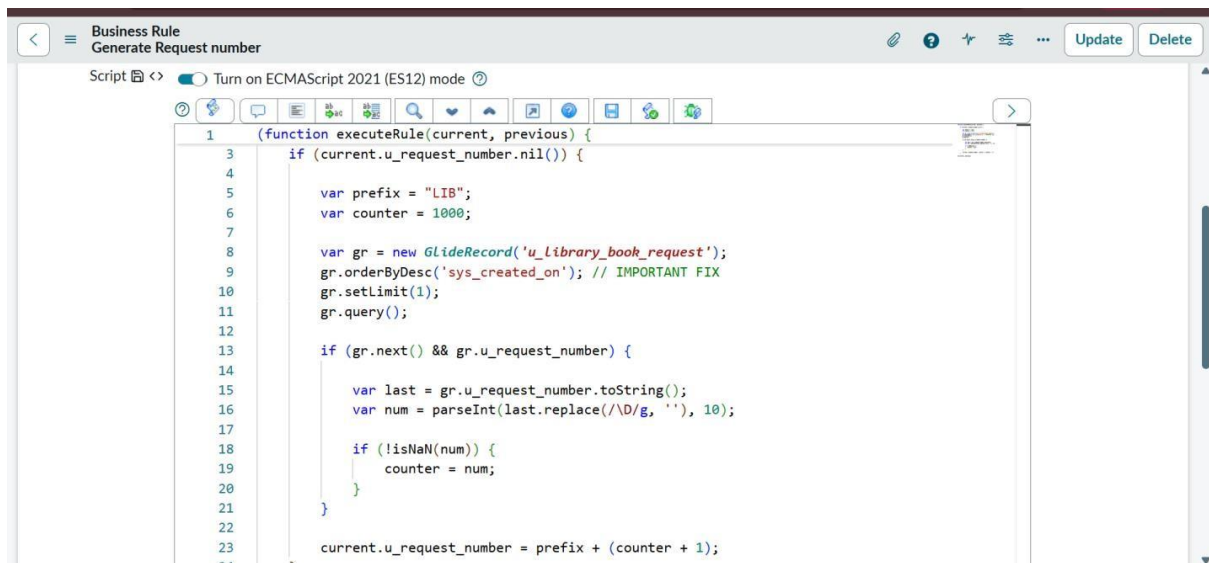
- LIB1001
- LIB1002
- LIB1003

This removes the need for manual entry and ensures that each request has a unique identifier.

Steps:

1. Navigate to **System Definition** → **Business Rules**.
2. Click **New**.
3. Enter the Business Rule name as **Generate Request Number**.
4. Select the **Library Book Request** table.
5. Set **When = Before** and check **Insert**.
6. Paste the Business Rule script.

7. Save the Business Rule.



Screenshot 4: Validate Issue Date Business Rule

Creating the Issue Date Validation Business Rule

A Business Rule was created to ensure that students cannot select an issue date that is earlier than the current date.

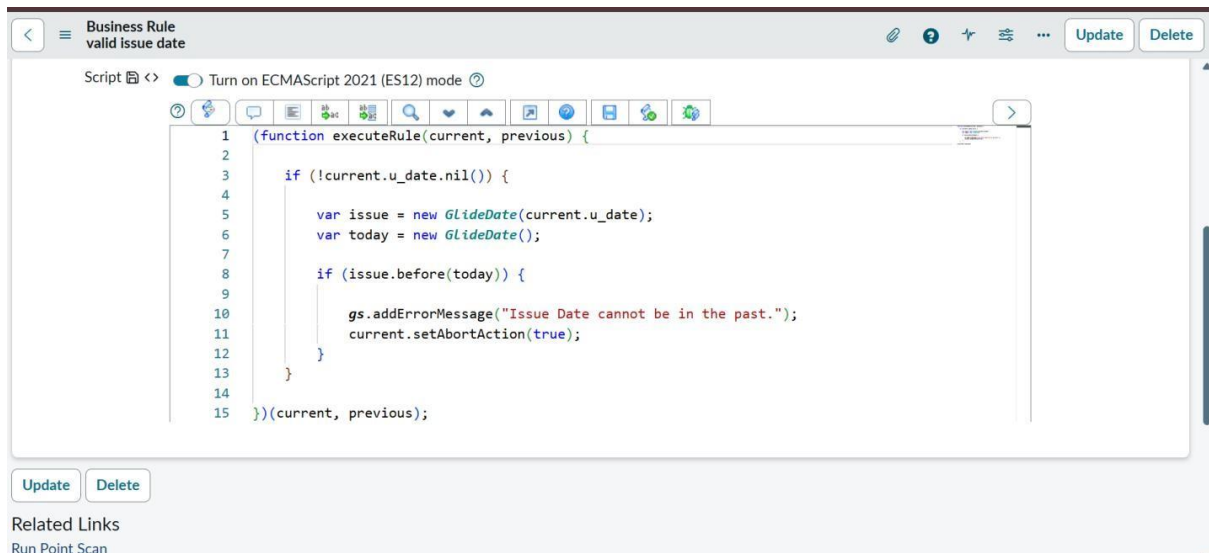
If a past date is selected, the system displays the following message:

Issue Date cannot be in the past.

This ensures data accuracy and prevents invalid requests.

Steps:

1. Navigate to **System Definition** → **Business Rules**.
2. Click **New**.
3. Name the rule **Validate Issue Date**.
4. Select the **Library Book Request** table.
5. Set **When = Before** and check **Insert**.
6. Paste the validation script.
7. Save the Business Rule.



Screenshot 5: Email Notification Configuration

Configuring Email Notification

An email notification was created to inform the librarian whenever a new book request is submitted.

The notification includes:

- Request Number
- Student Name
- Book Name
- Issue Date
- Status

Steps:

1. Navigate to **System Notification** → **Email** → **Notifications**.
2. Click **New**.
3. Enter the notification name.
4. Select the **Library Book Request** table.
5. Configure the notification to trigger on record insertion.
6. Add the recipient.
7. Enter the subject and HTML email body.
8. Save the notification.

Notification [sysevent_email_action]
Notify Librarian

Name: Notify Librarian

* Table: Library Book Request [u_library_bo...]

Mandatory: ☐

* Category: Uncategorized

Application: Global

Active: ☒

A digest combines and summarises multiple updates for a single or multiple target records within a set time interval.

Allow Digest: ☐

When to send | Who will receive | **What it will contain**

When using an **Email Template**, the **Subject** and **Message** content is picked from that template unless a different **Subject** or **Message** content is specified in this form.

Email template <>: Unsubscribe and Preferences

Subject: New Library Book Request Submitted

Screenshot 6: Successful Record Creation

Creating a Library Book Request

After completing the configuration, a new book request was created successfully.

The system automatically:

- Generated the Request Number
- Saved the record
- Stored the Issue Date
- Triggered the email notification

This confirms that all configured components are working correctly.

Emails

Created

Search

Actions on selected rows...

New

All

☐	Created	Recipients	Subject	Type	Notification type	User ID
	2026-05-15 01:00:09	aileen.mottern@example.com	Restocking Request For APC 42U 3100 SP2 NetShelter Rack	send-ready	SMTP	(empty)
	2026-04-29 22:22:11	admin@example.com	[PA VSC] Daily Data Collection completed with error	send-ready	SMTP	(empty)
	2026-06-18 01:00:10	aileen.mottern@example.com	Restocking Request For Fujitsu 1TB Hybrid Solid State Drive	send-ready	SMTP	(empty)
	2026-06-29 00:00:14	admin@example.com	Daily job to fetch Email Indicator Data and Email Notifications created completed with error	send-ready	SMTP	(empty)
☒	2026-06-26 09:06:59	admin@example.com	New Library Book Request Submitted	send-ready	SMTP	(empty)
	2026-05-20 00:00:13	admin@example.com	Daily job to fetch Email Indicator Data and Email Notifications created completed with error	send-ready	SMTP	(empty)
	2026-06-29 21:02:01	ron.kettering@example.com	On-Call Schedule - Gaps and Conflicts	send-ready	SMTP	(empty)

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Screenshot 7: Validation Error (Past Date)

Testing the Issue Date Validation

The Issue Date validation was tested by selecting a date earlier than the current date.

The Business Rule prevented the record from being saved and displayed the following error:

Issue Date cannot be in the past.

This confirms that the validation works correctly.

⊗ Issue Date cannot be in the past.

⊗ Invalid insert

Request number

LIB1001

Student name

Abraham Lincoln

ⓘ

Book name

Java Programming

Author

James Gosling

Issued date

2026-06-25 02:09:59

📅

Status

On Hold

8. Source Code

Include the following Business Rules:

Business Rule 1

```

1 (function executeRule(current, previous){
2
3     if (current.u_request_number.nil()){
4
5         var prefix = "LIB";
6         var counter = 1000;
7
8         var gr = new GlideRecord('u_library_book_request');
9         gr.orderByDesc('sys_created_on'); // IMPORTANT FIX
10        gr.setLimit(1);
11        gr.query();
12
13        if (gr.next() && gr.u_request_number){
14
15            var last = gr.u_request_number.toString();
16            var num = parseInt(last.replace(/\D/g, ''), 10);
17
18            if (!isNaN(num)) {
19                counter = num;
20            }
21        }
22
23        current.u_request_number = prefix + (counter + 1);
24    }
25}

```

Generate Request Number

Business Rule 2

Validate Issue Date

Script Editor Turn on ECMAScript 2021 (ES12) mode

```

1 (function executeRule(current, previous) {
2
3     if (!current.u_date.nil()) {
4
5         var issue = new GlideDate(current.u_date);
6         var today = new GlideDate();
7
8         if (issue.before(today)) {
9
10            gs.addErrorMessage("Issue Date cannot be in the past.");
11            current.setAbortAction(true);
12        }
13    }
14
15 })(current, previous);

```

9. Conclusion

The **Library Book Request Management System** successfully automates the library request process using the ServiceNow platform. The application minimizes manual effort by validating issue dates, automatically generating request numbers, and notifying the librarian whenever a new request is submitted.

The project improves efficiency, ensures accurate record management, and demonstrates the practical implementation of ServiceNow features such as Custom Tables, Forms, Business Rules, and Email Notifications.